DX-SMART

DX-LR02-900T22D
Serial port application guide

Version: 2.0

Date：2025-11-17

深圳大鼎光电科技有限公司

Updated records

Version	Date	Instructions	Author
V1.0	2024/03/09	Initial version	SML
V1.1	2024/07/22	Optimization instructions	SML
V2.0	2025/11/17	Modifying parameter units Update module usage action example New Instructions Modify instruction description and remarks	SML

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1. The introduction

DX - LR02-900T22D is a LoRa module, low power consumption is summer in SHENZHEN DX-SMART technology co., LTD. For intelligent wireless data transmission, domestic ASR6601 SOC chip, Chip internal integrated rf transceiver, the SUb 1GHZ Arm China STAR - MC1 microprocessor, built-in Flash storage, SRAM. This module supports UART, I2C, I2S interface, support IO port control, ADC acquisition, has the advantages of low power consumption, high performance and long distance. Applicable to a variety of applications in the field of IoT scenarios, such as smart meter, intelligent logistics, intelligent building, city of wisdom, and wisdom agriculture, and many other scenarios.

1.1. A serial port basic parameters

- Default parameters of module serial port: 9600bps/8/n/1 (baud rate/data bit/no check/stop bit)

1.2. Module rf basic parameters by default

- Module rate level: LEVEL2 stores (2148 BPS)
- Switching module mode: the way a (the AT command control)
- Module subpacket length: 230 bytes
- Packet module RSSI: closed
- Module key: 12345
- Module frequency band: 915.15MHz
- Transmit power of module: 22dBm
- Module working mode: transparent transmission
- Module power consumption modes: high aging model
- Module address: ffff
- Module CAD detection time: 4s

1.3. Transmission mode and AT command mode

- Transfer mode: module after power on, is the transmission mode, can begin to transmit data at this time.
- The AT command mode: in transmission mode, the use of "+ + +" switch to the AT command mode, can respond to the AT command. To enter the transmission mode, you need to send "+++" to exit the AT command mode.

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2. PC side test tool

2.1. Pc-side testing software

PC test software in the data package, please download and install sscom5.13.1 test, computer serial port software serial port software interface is as follows:



Figure 1: Computer side serial port software diagram

3. A serial port using

3.1. Module test minimum system

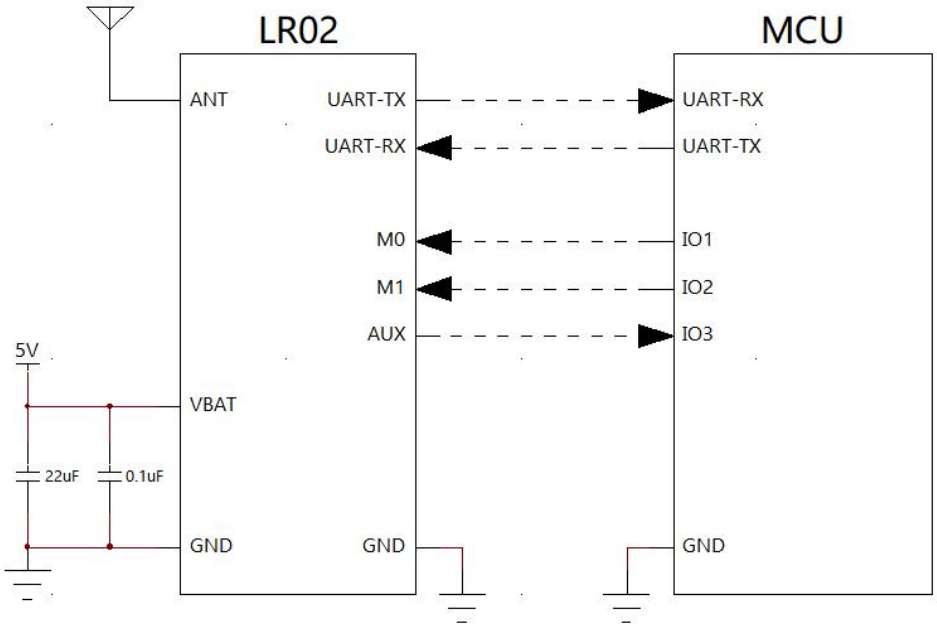


Figure 2: the module's minimum system diagram

3.2. Module USES matters needing attention

- For LoRa modulation mode, the characteristics of information transmission delay longer, suggest customers not under low airspeed for large amount of data transmission, in order to avoid data loss caused by the accumulated data communication exception.
- When the module from the other mode is switched to sleep mode, if there are data has not been processed, the module will these data (including received and sent) after processing, can enter into sleep mode.
- When in sleep mode into the other mode, the module will start configuration parameters, the configuration process, AUX maintain a high level; After the completion of the output low level, so suggest user testing AUX level changes.

3.3. Module USES operation example

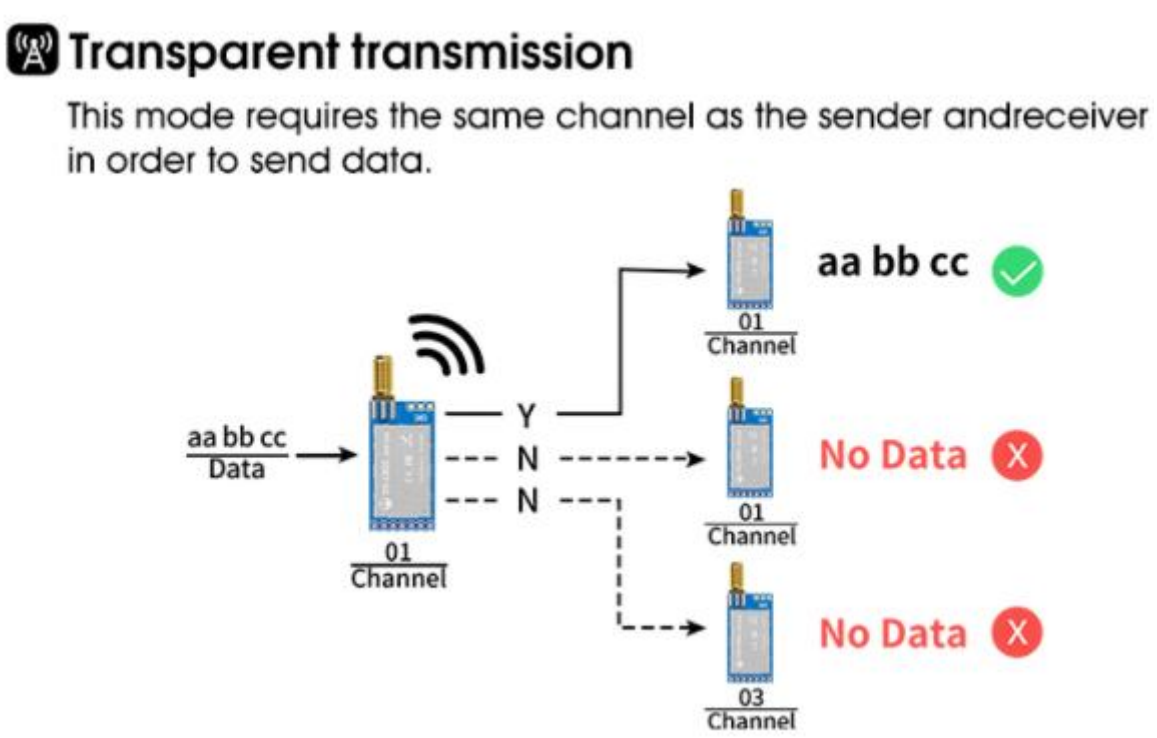
3.3.1. Transparent transmission module and module

With two DX - LR02-900 t22d module as an example, the module a and module b respectively

1. A and b connected a serial port and power supply module
2. Module a and b into the AT mode: + + +
3. Module a and b is set to transparent transmission mode: the AT + MODE0
4. Module a and b level of configuration the same rate: the AT + LEVEL2 stores
5. Module a and b: configuration of the same channel AT + CHANNEL01
6. Module a and b to restart command takes effect: the AT + RESET
7. Data transfer: a module to send data, another module can receive data

Note: (1) lora is half-duplex protocol, so only a module to send at a time

(2) Transparent transmission mode requires the same rate level and channel of two modules to transmit data to each other



3.3.2. Fixed transmission module and module

Two DX-LR02-900T22D modules, module a and module b, are taken as an example

1. A and b connected a serial port and power supply module

2. Module a and b into the AT mode: + + +
3. Module a and b is set to the fixed-point transport pattern: the AT + MODE1
4. Modules a and b are configured with the same rate level: AT+LEVEL2
5. Module a and b restart command takes effect: AT+RESET
6. Data transfer: two module choice HEX format to send, and according to the specific data format data should be transmitted between them

Note: (1) Flora is half-duplex protocol, so only a module to send at a time

(2) the fixed-point transport pattern belong to private agreement, need two modules rate of the same grade, and according to the following data format, to send and receive data properly:

Data format: the address of the receiver (hexadecimal, two bytes) + the receiver channel (hexadecimal, 1 byte) + data (hexadecimal)

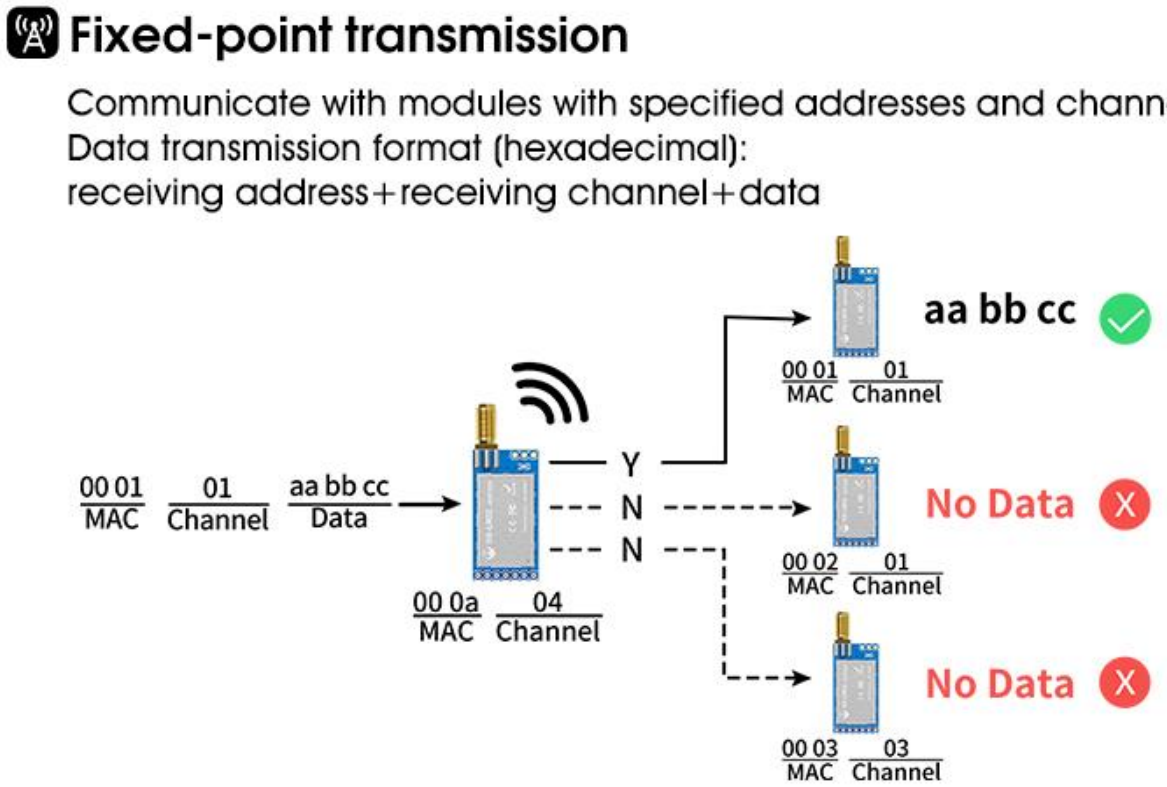
For example:

(module as the sending end; a module b as the receiver, the module b address is 0001, channel 01)

Module to send data to the module b aabbcc:

(1) Module a selects HEX format to send

(2) Module a send data content for: 00 01 01, 61, 61, 62, 62, 63, 63



3.3.3. Radio transmission module and module

Two DX-LR02-900T22D modules, module a and module b, are taken as an example

1. A and b connected a serial port and power supply module
2. Module a and b enter AT mode: +++
3. Modules a and b are set to broadcast transmission mode: AT+MODE2
4. Module a and b level of configuration the same rate: the AT + LEVEL2 stores
5. Module a and b restart command takes effect: AT+RESET
6. Data transfer: two module choice HEX format to send, and according to the specific data format data should be transmitted between them

Note: (1) lora is a half-duplex protocol, so only one module can be sent at a time

(2) The broadcast transmission mode is a private protocol, which requires the same rate level of the two modules and the following data format in order to receive and send data normally:

Data format: receiver channel (hexadecimal, 1 byte) + data (hexadecimal)

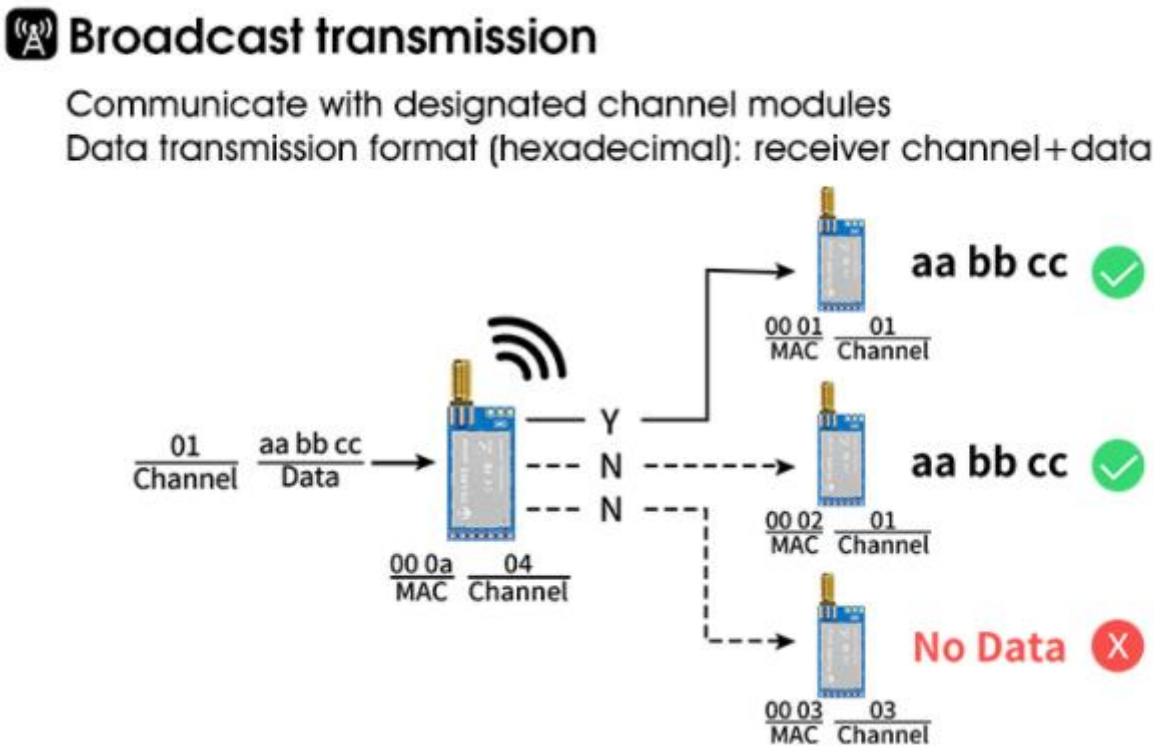
For example:

(module as the sending end; a module b as the receiver, the module b channel 01)

Module a sends data aabbcc to module b:

(1) Module a selects HEX format to send

(2) Module a sends the data content as: 01 61 61 62 62 63 63



3.3.4. Mode switching

3.3.4.1. Mode 1 (Configure operation mode via AT command)

+++ Enter the AT command mode and configure the desired mode with the following command:

- High Aging mode: AT+SLEEP2
- Air Wake Mode: AT+SLEEP1
- Sleep mode: AT+SLEEP0

3.3.4.2. Way 2 (by M0 / M1 pin configuration four work modes)

When the module command AT + SWITCH = 1, by M0 / M1 pin SWITCH module working mode, working pattern corresponding to the table below:

M0 input level	M1 input level	Operating mode
High level	High level	Sleep mode
High level	Low level	Air wake-up mode
Low power level	High level	AT mode
Low level	Low level	High aging mode

Notes

1、Command AT + SWITCH details please refer to 5.3.5 Settings \ query - hardware control pin.
2、The command AT + SWITCH = 1, M0 / M1 pins for internal weak pull-up, by M0 / M1 pin SWITCH module working mode, the impending, please be sure to avoid pin to prevent abnormal mode switches.

3.3.5. AUX module status indicates the foot work

Pin name	Pin number	Description	Pin output level	Instructions
AUX	5	Module working status indication	High level	Data transmitting/data receiving/working mode switching
			Low level	Data send complete/data receive complete/work mode switch complete

note

When the module detects the received data, AUX will output a high level 2-3ms in advance, which is used to prompt the external MCU to be ready to receive the data.

3.3.6. Remote control M0 / M1 pins

When the module command AT + SWITCH = 0, module receives the following parameters, will change the high and low level of M0 / M1 pin, level corresponding to the table below:

Receive parameters	M0 output level	M1 output level
d10x	Low level	-
d11x	High level	-
d20x	-	Low level
d21x	-	High level

Notes

1, this state is only valid if both parties are AT+OPENKEY is 1.
2, send parameters, need not add carriage returns.

4. Related AT command details

4.1. Command Format Description

The AT + Command < param1, param2, param3 > < CR > < LF >

- All commands begin with AT and end with <CR><LF>. In the table that shows commands and responses in this document, <CR><LF> is omitted and only commands and responses are displayed.
- All the AT command in English uppercase characters.
- <> is optional content, if there are multiple arguments in the command, separated by a comma ", ", the actual command does not contain Angle brackets.
- <CR> is the carriage return character \r, which is 0X0D in hexadecimal.
- < LF > for newline characters \ n, hex 0 x0a.
- Instruction execution is successful, returns the corresponding command ends with a OK, failure return EEROR = < >, "< >" content for the corresponding error code (see 5.4).

4.2. Response Format Notes

+Indication<=param1, param2, param3><CR><LF>

- Response instructions with a plus sign "+", < CR > < LF > end
- Is equal to "=" in response to the parameters
- If there are multiple arguments in the response argument, they are separated by a comma ","

4.3. The AT command for example

Example: modify LoRa equipment baud rate to 9600

Send: AT+BAUD3

Return: OK

4.4. The AT command list

Instructions	function	instructions
+++	To enter or exit the AT command mode	Power on defaults to transmission mode
AT	Test instructions	Used to test the serial port
AT+RESET	Software restart	-
AT+DEFAULT	factory data reset	-
AT+BAUD	Set \ Query serial port baud rate	Default: 3 (9600)
AT+PARI	Set \ Query serial port check bits	Default: 0 (no parity)
AT+HELP	Query configuration information	-
AT+LEVEL	Set \ Query module air rate and communication distance	Default: 2
AT+MODE	Set \ Query transmission mode	Default: 0 (transparent transfer)
AT+SLEEP	Set \ Query work mode	Default: 2 (high aging model)
AT+SWITCH	Set \ Query hardware control pin status	Default: 0 (off)
AT+CHANNEL	Set \ Query working channel	Default: 41
AT+MAC	Set \ Query device address	Default: ff,ff
AT+OPENKEY	Set \ Query module key switch	Default: 1 (on)
AT+KEY	Set the key module	Default: 12345
AT+PACKET	Set \ query subpacket length	Default: 3 (230 bytes)
AT+DRSSI	Set \ query packets RSSI	Default: 0 (closed)
AT+POWE	Set \ Query transmit power	Default: 22
AT+LBT	Set \ Query LBT status	Default: 0 (off)
AT+LRSSI	Set \ query LBT listening threshold	Default: -87
AT+ERSSI	Query the current channel noise level	-

5. AT command details

5.1. Basic commands

5.1.1. Enter or exit the AT command mode

Features	Instructions	Response	Instructions
Enter or exit the AT command mode	+++	Entry AT or Exit AT Power On	Entry AT: Enter the AT command mode Exit AT: Exit the AT command mode Power on defaults to transfer mode

Remark:

1、It will be reset automatically when you exit the AT command mode.
2、This instruction is not saved when power supply drop.

5.1.2. Test the instruction

Function	Instructions	Response	Instructions
Testing	AT	OK	

5.1.3. Software restart

Features	Instructions	Response	Instructions
Software restart	AT+RESET	OK Power On	

5.1.4. factory data reset

Features	Instructions	Response	Instructions
factory data reset	AT+DEFAULT	OK Power On	

5.2. Serial port parameters

5.2.1. Settings \ Query - Serial Baud rate

Features	Instructions	Response	Instructions
Query baud rate	AT+BAUD	+BAUD=<baud>	< baud > baud rate corresponding to the serial number 1: 2400 5:38,400 2:4800 6:57600 3:9,600 7:115200 4:19200 Default: 3(9600)
Set baud rate	AT+BAUD<baud>	+BAUD=<baud> OK	

Notes:

After setting this instruction, it should be restarted to take effect.

5.2.2. Set \ Query - serial port check bit

Features	Instructions	Response	Instructions
Query check digit serial port	AT+PARI	+PARI=<param>	< param> sequence number 0: No validation 1: odd check 2: Even check Default: 0
Set the serial port check bit	AT+PARI<param>	+PARI=<param> OK	

Notes:

After setting this instruction, it should be restarted to take effect.

5.3. LORA parameters

5.3.1. Query configuration information

function	Instructions	Response	Instructions
Query basic module configuration information	AT+HELP	========== LoRa Parameter: +VERSION=<version> MODE:<mode> LEVEL:<level> SLEEP:<sleep> Frequency:<frequency> MAC:<mac> Bandwidth:<bandwidth> Spreading Factor:<spreading factor> Coding rate:<coding rate> CRC:<crc> IQ:<iq> Power:<power> ==========	LoRa Parameter: LoRa parameter <version>: The version <mode> : Transmission mode < level > : air rate configuration <sleep> : Working mode The < frequency > : operating frequency <mac> : device address <bandwidth> : RF bandwidth The spreading factor > : spreading factor < coding rate > : radio frequency coding rate "CRC > : CRC check <iq> : Whether the IQ signal is flipped <power> : Transmit power

For example:

Query module basic information

Send: AT+HELP

Return to: ==========

LoRa Parameter:

+ VERSION = V3.1.5

```
MODE:0
LEVEL:2 >> 2148bps
SLEEP:2
Frequency:915150000hz >> 41
MAC:ff,ff
Bandwidth:2
Spreading Factor:11
Coding rate:1
CRC:0(false)
IQ:0(false)
Power:22dBm
====================================
```

5.3.2. Settings \ query - a key air rate configuration module and communication distance

Features	Instructions	Response	Instructions
Query module parameters	AT+LEVEL	+LEVEL =<param>	<param> : 0-7, Air rate and the communication distance configuration, there are eight gear Default value: 2
Set the module parameters	AT+LEVEL<param>	+LEVEL =<param> OK	

Notes:

1, can choose according to their own communication distance and quantity of the data of different gear (data volume and distance can refer to the table below). Air rate, the greater the characters can be sent the amount of data the faster, the shorter the distance communication. The air rate is inversely proportional to the distance characters.

2, the instruction will be radio frequency bandwidth, radio frequency coding rate, spreading factor are set up, and can be used directly.

3, emission and receiving equipment LEVEL gear should be consistent to send and receive data.

4. After setting the instruction, it should be restarted to take effect.

Note: the following table for the configuration parameters under different gear, the following open visual range, for reference only, actual distance will be subject to the measured.

LEVEL (gear)	SF(spread spectrum factor)	BW (KHz bandwidth)	CR (coding rate)	Air character rate (bit/s)	Open visible distance (Km)
0	11	125	4/8	336	8.0
1	11	250	4/5	1075	-
2	11	500	4/5	2148	-
3	8	250	4/5	6250	-
4	8	500	4/6	10417	-
5	7	500	4/6	18229	2.4
6	6	500	4/5	37500	-
7	5	500	4/5	62500	-

5.3.3. Set \ query - transfer mode

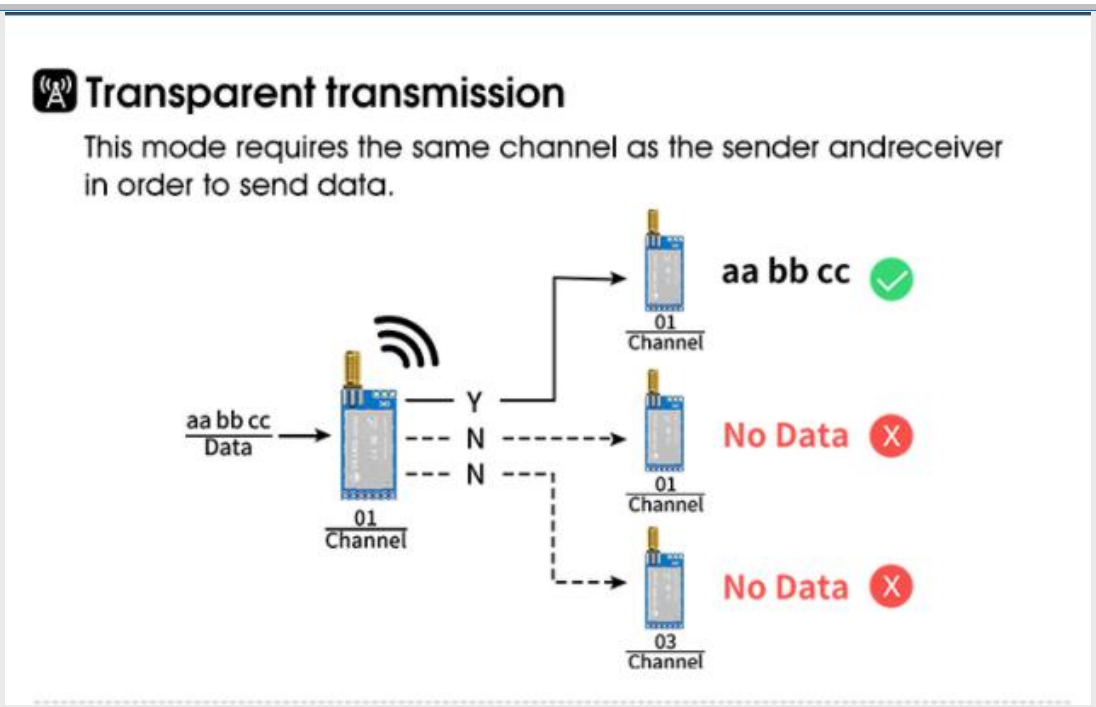
Features	Instructions	Response	Instructions
Query transmission mode	AT+MODE	+MODE=<param>	param: 0,1,2 0: Transparent transmission 1: Fixed-point transmission 2: Broadcast transmission Default setting: 0
Set transmission mode	AT+MODE<param>	+MODE=<param> OK	

Notes:

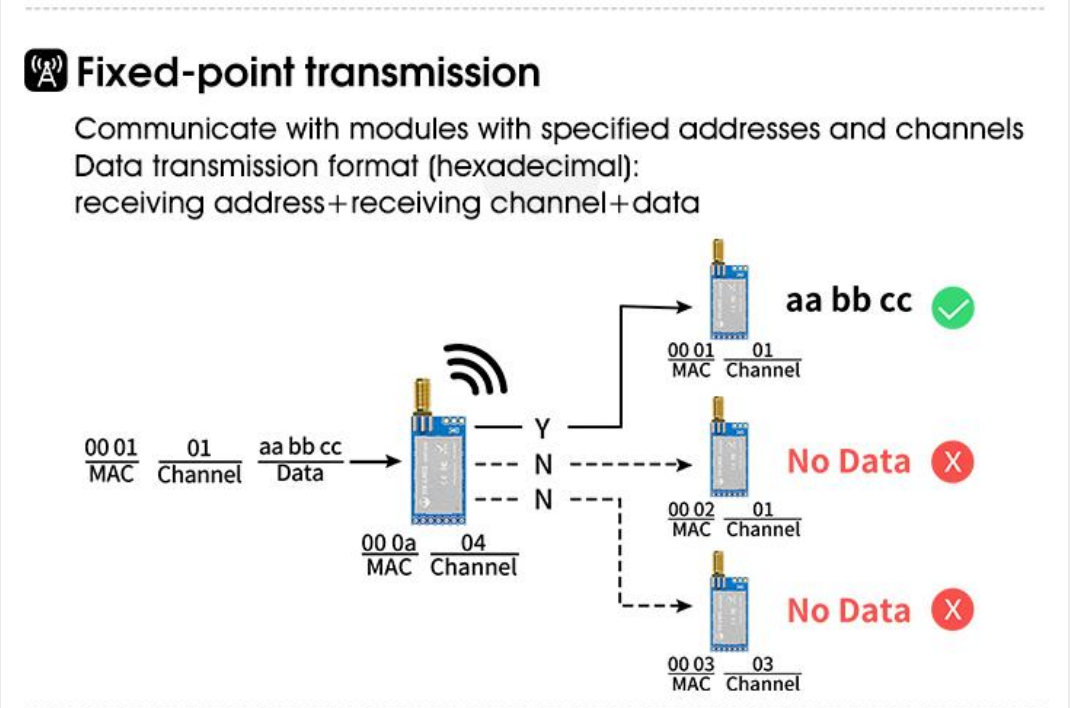
1、After setting this instruction, it should be restarted to take effect.
2、Transparent transmission data format: Send data directly
3、Fixed-point transmission data format: device address (hexadecimal, two bytes) + channel number (hexadecimal, one byte) + data (hexadecimal)
4. Broadcast transmission data format: channel number (hexadecimal, one byte) + data (hexadecimal)

Examples:

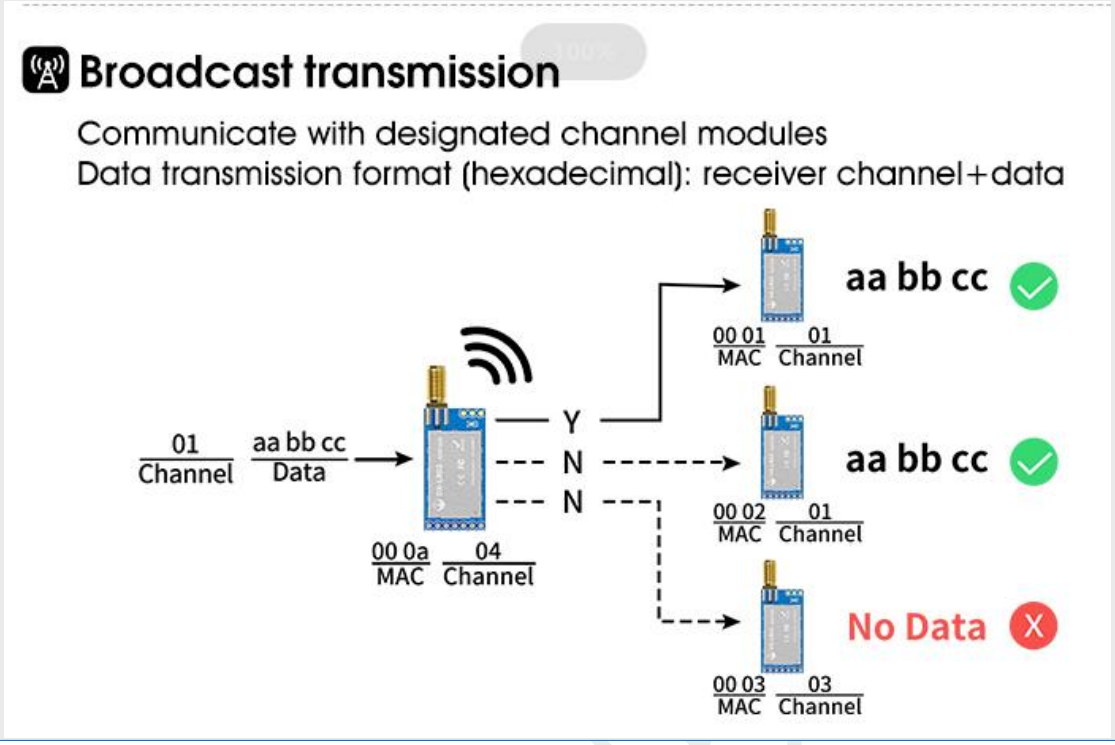
1、Transparent transfer: Sending data directly



2、Fixed-point transmission: the address of the receiving module is 0001, and the channel is 01; Emission module to send data to aabbcc, then send the data content is: 000101 aabbcc (hex: 00 01 01, 61, 61, 62, 62, 63, 63)



3、Radio transmission: receiving module for channel 01, emission module to send data to aabbcc, sending data content as: 01 aabbcc (hex: 01, 61, 61, 62, 62, 63, 63)



5.3.4. Set \ Query - working mode

Features	Instructions	Response	Instructions
Query working mode	AT+SLEEP	+SLEEP=<param>	< param> serial number Zero: sleep mode 1: Air wake-up mode 2: high aging model Default: 2
Set the work mode	AT+SLEEP<param>	+SLEEP=<param> OK	

Notes:

1、Sleep mode: In this mode, both MCU and RF go to sleep.

Method one: use a serial port to sleep and wake up, send the AT + SLEEP0 into sleep mode, wake up when sending to awaken AT + WAKEUP or any data, wake up after the module is in a state of the AT command.

Method 2: use, M0, M1 feet to sleep and wake up, send the AT + SWITCH1 open M0, M1 mode switch, module automatically restart, = 1 set M0, M1 = 1, enter the sleep mode, M0, M1, of arbitrary switching module out of sleep mode.

2、Air wake mode:

A. In this mode, the module performs CAD detection in a cycle of four seconds (the overall

sleep time is: 4s minus CAD detection time). If the module detects data, it will enter the receiving mode, and automatically enter the sleep after receiving data. Rf dormancy during dormancy, MCU don't sleep.

B, using air wake model, the receiver and the sender should wake up in the air mode, can send and receive data.

C, the model can be written to save.

Note: CAD explanation: LoRa CAD (Channel Activity Detection) is used to detect in LoRa network Channel Activity of a kind of technology. It is used to determine whether there is activity (such as transmission by other devices) on a specified physical channel to help the device choose the right time to send and avoid collisions.

3、High aging mode: the mode, the module has been at the receiving state, ready to receive the data of other equipment. When the module of serial port to receive the master data, namely state, switch to launch the data launch out, after the completion of the launch, switch back to the receiving state.

4、Set the module takes effect immediately after sleep mode.

5、After the rest of the two mode need to restart to take effect.

5.3.5. Settings \ Query - Hardware control pin status

Features	Instructions	Response	Instructions
Query the hardware control pin status	AT+SWITCH	+SWITCH=<param>	<param> : 0,1 0: close 1: Open Default: 0 (off)
Sets the hardware control pin state	AT+SWITCH<param>	+SWITCH=<param> OK	

Notes:

1、This command automatically restarts.

2、When the SWITCH = 1, through the module hardware pins M0, M1 to select the work mode: (the following 0 as the low level, 1 is the high level)

M0 = 0, M1 equals zero, the module in high aging model;

When M0=1, M1=0, the module is in the air wake-up mode;

M0 = 0, M1 = 1, the module in the AT mode;

When M0=1, M1=1, the module is in sleep mode.

3、Through pin control into dormancy, through a serial port or instruction, must be out of

hibernation by M0, M1.

4、When the SWITCH = 0, can pass the AT + SLEEP this instruction to select the work mode.

5.3.6. Channel Settings \ query - work

function	Instructions	Response	Instructions
Query the working channel	AT+CHANNEL	+ CHANNEL=<param>	param: 00-63 (hexadecimal) Starting at 850.15Mhz, The channel interval is 1Mhz The default Settings: 41
Set up working channel	AT+CHANNEL <param>	+CHANNEL=<param> OK	

Notes:

1, set up a 100 general channel, this module can contact contact us for more.

2. After the instruction is set, it needs to be restarted.

3. When multiple receiving devices are too close to the transmitting device, it may lead to the receiving device of different channels can receive data, so the distance between the transmitting device and the receiving device is required to be as far as possible.

Note: the following table for different channel of the frequency control, unit: Mhz.

channel	Operating frequency band	Channel	Operating frequency band	Channel	Operating frequency band	channel	Operating frequency band
00	850.15	19	875.15	32	900.15	4B	925.15
01	851.15	1A	876.15	33	901.15	4C	926.15
02	852.15	1B	877.15	34	902.15	4D	927.15
03	853.15	1C	878.15	35	903.15	4E	928.15
04	854.15	1D	879.15	36	904.15	4F	929.15
05	855.15	1E	880.15	37	905.15	50	930.15
06	856.15	1F	881.15	38	906.15	51	931.15
07	857.15	20	882.15	39	907.15	52	932.15
08	858.15	21	883.15	3A	908.15	53	933.15
09	859.15	22	884.15	3B	909.15	54	934.15

0A	860.15	23	885.15	3C	910.15	55	935.15
0B	861.15	24	886.15	3D	911.15	56	936.15
0C	862.15	25	887.15	3E	912.15	57	937.15
0D	863.15	26	888.15	3F	913.15	58	938.15
0E	864.15	27	889.15	40	914.15	59	939.15
0F	865.15	28	890.15	41	915.15	5A	940.15
10	866.15	29	891.15	42	916.15	5B	941.15
11	867.15	2A	892.15	43	917.15	5C	942.15
12	868.15	2B	893.15	44	918.15	5D	943.15
13	869.15	2C	894.15	45	919.15	5E	944.15
14	870.15	2D	895.15	46	920.15	5F	945.15
15	871.15	2E	896.15	47	921.15	60	946.15
16	872.15	2F	897.15	48	922.15	61	947.15
17	873.15	30	898.15	49	923.15	62	948.15
18	874.15	31	899.15	4A	924.15	63	949.15

5.3.7. Settings \ Query - Device Address

Features	Instructions	Response	Instructions
Looking up device addresses	AT+MAC	+MAC=<param>,<param>	param: Hexadecimal, one byte The default Settings: ff ff
Set device address	AT+MAC<param>,<param>	+MAC=<param>,<param> OK	

Notes:

After setting this instruction, it should be restarted to take effect.

Examples:

The module address is set to 0 a01
Send: AT+MAC0a,01
Return: +MAC=0a,01
OK

5.3.8. Settings \ Query - Module Key switch

Features	Instructions	Response	Instructions
Query module is the key switch	AT+OPENKEY	+OPENKEY=<param>	<param> 0,1 0: Close key 1: Turn key on Default value: 1
The key switch setting module	AT+OPENKEY<param>	+OPENKEY=<param> OK	

Notes:

1、After setting this instruction, it should be restarted to take effect.
2、The AT + OPENKEY = 0, can turn off the key function, but are not clear set of keys.
3、The OPENKEY of both sender and receiver must be consistent, otherwise the sender and receiver will be abnormal.

5.3.9. Setup - Module key

Features	Instructions	Response	Instructions
Set the module key	AT+KEY<param>	+KEY=<param> OK	<param>0~65535 Default value: 12345

Notes:

1. After setting the instruction, it should be restarted to take effect.
2. The instruction can only be set and cannot be queried.
3. The key of both sending and receiving must be consistent, otherwise the sending and receiving will be abnormal.
4. If you need to restore the key value to the DEFAULT, you need to send AT+DEFAULT to restore all the module parameters to the default value.

5.3.10. Set \ query-subpacket length

Features	Instructions	Response	Instructions
Inquire about the subcontract length	AT+PACKET	+ PACKET=<param>	param: 0 to 3 0:32 bytes 1:64 bytes 2:128 bytes 3:230 bytes
Set the	AT+PACKET<param>	+PACKET=<param>	Default: 3 (230 bytes)

subpacket length	OK

Notes:

After setting this instruction, it should be restarted to take effect.

5.3.11. Settings \ Query - Packet RSSI

Features	Instructions	Response	Instructions
Query packet RSSI	AT+DRSSI	+DRSSI=<param>	<param> : 0,1 0: close 1: Open Default: 0 (off)
Set packet RSSI	AT+DRSSI<param>	+DRSSI=<param> OK	

Notes:

1、After setting this instruction, it should be restarted to take effect.
2、The received signal strength is added to the received packet.

Examples:

Sender: 31 32 33 34 35
Receiver: 31 32 33 34 35 AB
Where the last bit "AB" is the RSSI value detected in real time, which will change with the strength of the signal.
The RSSI value is calculated as follows (using the value AB as an example) :
Actual signal strength =- (0xFF-0xAB) =- (255-171) =-84 dBm
Therefore, the received signal strength for this time is -84 dBm. Note that the RSSI value may be different for each communication, and it depends on the current wireless signal quality. 0xFF in the above formula is a fixed value.

5.3.12. Set \ Query - Transmit power

Features	Instructions	Response	Instructions
Query transmit power	AT+POWE	+POWE=<param>	param: 0-22dBm (integer

Set transmit power	AT+POWE<param>	+POWE=<param> OK	value) Default setting: 22dBm

Notes:

After setting this instruction, it should be restarted to take effect.

Examples:

Modify the transmit power to 10dBm

Send: AT+POWE10

Return: +POWE=10

OK

5.3.13. Set \ query -LBT status

Features	Instructions	Response	Instructions
Check LBT status	AT+LBT	+LBT=<param>	<param> : Sequence number 0: closed 1: Open Default: 0
Set the LBT status	AT+LBT<param>	+LBT=<param> OK	

Notes:

1、After setting this instruction, it should be restarted to take effect.

2、When the LBT is set to 1, the channel noise will be monitored first. When the noise is greater than the threshold, the channel noise will be monitored continuously, and it will be sent directly after 2S.

5.3.14. Set the \ query-lbt listening threshold

Features	Instructions	Response	Instructions
Query the LBT listening threshold	AT+LRSSI	+LRSSI=<param>	<param> : -255 to 0 The current LBT listening threshold Default value: -87
Set the LBT listening threshold	AT+LRSSI<param>	+LRSSI=<param> OK	

Notes:

1、After the instruction is set, it needs to be restarted to take effect.
2、Changing the parameter will change the LBT listening threshold.
3. This default value is obtained from our company's test in a specific environment, and the user needs to set the noise level when the channel is idle through AT+ERSSI.

5.3.15. Query - current channel noise level

Features	Instructions	Response	Instructions
Query the current channel noise level	AT+ERSSI	+ERSSI= <param>	<param> : Current channel noise level

Remarks:

Enquiries only.

5.4. List of error codes

The details of the error code in EEROR=<> are as follows:

Return value	Error Message Description
104	Invalid instructions
105	Invalid parameters
106	Other errors

6. Value-added services

In order to meet the various functional requirements of customers, our company can provide the following technical value-added services:

- Module program customization, such as: IO function port customization, AT instruction customization, etc.
- Module PCB hardware customization, can be customized to the hardware requirements of customer needs.
- A variety of program customization, can be customized according to customer needs, a full set of software and hardware solutions.
- A full set of networking solutions customization, can be customized according to customer needs, a full set of networking, gateway solutions.

If you have the above customized requirements, please contact our business personnel directly.

深圳大夏龙雀科技有限公司